

## 4.5 Data Visualization & Types of Graphs

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیتھر دو متغیر...

CS XII

Chapter 4: Data and Analysis

Lecture 07 of 8

★ Golden

### 4.5 Data Visualization & Types of Graphs ★ Golden

#### LEARN IT

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

| Chart        | Shows                                               | Watch out                                                |
|--------------|-----------------------------------------------------|----------------------------------------------------------|
| Bar / column | Compare discrete categories (subjects, departments) | Column = vertical; bar often = horizontal                |
| Line         | Change over time (temperature, visitors)            | Needs an ordered x-axis (months, years)                  |
| Pie          | Parts of one whole as percentages                   | Bad with many or almost-equal slices                     |
| Histogram    | Distribution of one numeric variable in bins        | Bars <b>touch</b> (continuous range), unlike a bar chart |
| Scatter      | Relationship of two numeric variables               | Each dot is one pair (x, y)                              |
| Box plot     | Min, Q1, median, Q3, max, and outliers              | Best tool for spotting extreme values                    |

#### 4.5.1 Bar / column

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

#### 4.5.2 Line

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
temps = [12, 15, 20, 25, 30, 35]
plt.plot(months, temps, marker="o")
plt.title("Average Monthly Temperature")
plt.grid(True)
```

#### 4.5.3 Pie

```
subjects = ["Math", "Physics", "CS", "Biology"]
students = [25, 15, 30, 10] # total 80
plt.pie(students, labels=subjects, autopct="%1.1f%%", startangle=90)
```

Check: Math  $25/80 = 31.25\% \rightarrow 31.2\%$ . Physics  $18.8\%$ . CS  $37.5\%$ . Biology  $12.5\%$ .

#### 4.5.4 Histogram

```
scores = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 90, 95]
plt.hist(scores, bins=[40, 50, 60, 70, 80, 90, 100], edgecolor="black")
```

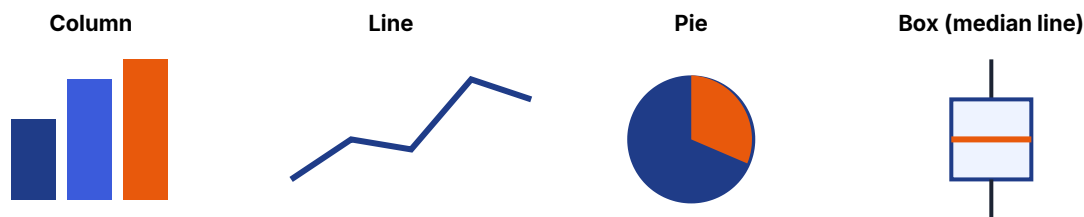
#### 4.5.5 Scatter

```
plt.scatter([10, 20, 30, 40, 50, 60, 70, 80],
            [2, 4, 5, 4, 7, 8, 9, 12], color="green", alpha=0.7)
plt.title("Rainfall vs Crop Yield") # positive correlation
```

#### 4.5.6 Box plot

```
plt.boxplot([[55, 60, 65, 70, 75, 80, 85],
            [40, 65, 70, 72, 75, 80, 95]],
            tick_labels=["Class A", "Class B"])
```

The box is the middle 50% (Q1 to Q3). The line inside is the median. Whiskers reach min and max (or a fence, with outliers drawn as dots). Class B is more spread out than Class A.



Column compares; line shows a trend; pie shows share; the box summarises spread around the median.

#### EXAM TIP

"Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

#### اردو میں سمجھیں

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکینڈ دو متغیرات کا تعلق، باکس پلاٹ پانچ اعداد اور آؤٹ لائر۔

#### CHECK YOURSELF

1. Why do histogram bars touch?
2. Rainfall vs yield — which chart?

Answers: 1. The x-axis is a continuous numeric range, not separate categories. 2. Scatter (two numeric variables, looking for correlation).

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

### MCQ 1. Bar/column charts compare:

- a. Discrete categories
- b. Parts of one whole only
- c. A single time point of one var as a distribution
- d. A 5-number summary

### MCQ 2. Line graphs are for:

- a. Trends over time
- b. Unrelated categories with gaps
- c. Shares of a budget
- d. Outliers in five numbers

### MCQ 3. Pie charts need:

- a. Parts of one whole; few slices
- b. 60 slices of 1%
- c. Time on x
- d. Two numeric axes of correlation

### MCQ 4. Histogram bars:

- a. Touch (continuous bins)
- b. Must have large gaps
- c. Are always time
- d. Replace the mean

### MCQ 5. Scatter plots show:

- a. Two numeric variables (correlation)
- b. One category vs colour
- c. FIFO order
- d. OSI layers

### MCQ 6. A box plot shows:

- a. Min, Q1, median, Q3, max (+ outliers)
- b. Only the mean
- c. Only mode
- d. Only n

## Extra short questions

1. One-line job of each of the six charts.
2. Why histogram  $\neq$  bar.
3. When pie fails.
4. Rainfall vs yield?

## Extra long question

1. Six school questions  $\rightarrow$  pick the chart and a matplotlib function (bar, plot, pie, hist, scatter, boxplot). Justify in one sentence each. ★ Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Bar categories; line time; pie share; hist distribution; scatter relationship; box spread/outliers.
2. Numeric bins, touching bars vs separate categories.
3. Too many / nearly equal slices.
4. Scatter; look for positive correlation.

### Long

1. Enrolment by faculty: bar. Hits by month: plot. Club shares: pie. Heights: hist. Hours vs marks: scatter. Two classes' spread: boxplot. Unlabelled charts score 0.